

Same Solution, Different Words: A Pre-Registered, Human-Audited Look at Pass@ k Shrinkage in a Small Language Model

Sidaarth Krishnakanth

Department of Computer Science
and Engineering
Sathyabama Institute of
Science and Technology
Chennai, India
Email: ksidaarth2005@gmail.com

Shivam Sunderam

Department of Computer Science
and Engineering
Sathyabama Institute of
Science and Technology
Chennai, India
Email: shivamsunderam.cse@gmail.com

Ms. T. Bhanu Shree

Department of Computer Science
and Engineering
Sathyabama Institute of
Science and Technology
Chennai, India
Email: ubhanukrithik@gmail.com

Abstract—Reinforcement learning with verifiable rewards (RLVR) raises the single-sample accuracy of language models, but whether it also shrinks the set of problems a model can solve at all is disputed: some studies find base models overtaking RLVR-trained ones at large k , others report expansion under longer training, and others trace the decline to how Pass@ k is measured. The evidence on all sides is automatic (Pass@ k , entropy, similarity metrics, LLM judges), so nobody has checked with people what RLVR actually loses; and fixes acting within a rollout group and across prompts have never been tested together. We address all three gaps on Qwen2.5-0.5B-Instruct trained with GRPO on MATH: a pre-registered 2×3 factorial crosses a macro factor (none vs. a topic-level budget on positive advantage mass) with a micro factor (none, the per-prompt SELF filter, or mean-preserving rarity redistribution over solution modes), with two or three seeds per cell, and a blinded human audit checks what is lost. At our budget of 2.0 rollouts per training problem, vanilla GRPO does not shrink the boundary: MATH-500 Pass@1 moves from 30.3% to 30.4% and Pass@32 rises from 73.0% to 75.3%, and every condition gains more problems than it loses. Neither pre-registered contrast is significant. Under the budget, replacing the SELF filter with mode-rarity redistribution changes the shrinkage slope by -0.02 pp per unit $\ln k$ (95% CI $[-0.79, +0.68]$, i.e. $[-2.7, +2.4]$ points of gain from $k = 1$ to $k = 32$), and adding the budget to the redistribution by $+0.22$ (CI $[-0.60, +1.05]$).^[R1] In the blinded audit, two annotators agreed on 85% of solution pairs (PABAK $+0.70$) and judged 50 of their 51 agreed pairs to use the same approach. The word-overlap partitioner that splits them into “modes” scores 55% against that consensus, below a trivial always-same labeller (98%), so the diversity these fixes reweight is mostly wording. The annotators’ labels of reasoning validity agreed no better than chance, leaving what was lost open.^[R17] We read the null as a matter of training intensity rather than model size. The protocol runs on one 24 GB GPU and is released.

Index Terms—reinforcement learning with verifiable rewards, GRPO, Pass@ k , reasoning boundary, small language models, credit assignment, human evaluation, pre-registration

I. INTRODUCTION

Group Relative Policy Optimization (GRPO) [1] and related RLVR methods have become the standard way to improve mathematical reasoning in open language models. Whether

they teach models to solve new problems, or only to solve solvable ones more reliably, is disputed. Yue et al. [2] found that base models overtake their RLVR-trained versions when many samples are drawn, a *Pass@ k inversion* since analysed as a shrinking reasoning boundary [3]–[6]. Others report the opposite: prolonged training expands the boundary [7], and RLVR comes out ahead when only correct chains of thought are counted [8]; a two-stage account reconciles the two by training duration [9]. A third view questions the measurement: aggregate declines partly reflect how few rollouts each training problem receives [10], and Pass@ k at large k rewards lucky hits [11]. The question matters most for small language models (SLMs), whose usefulness rests on sampling-based coverage (majority voting, verifier reranking) rather than on one greedy answer.

The evidence on every side is automatic. Pass@ k cannot tell a valid solution from a lucky guess; CoT-Pass@ k hands that judgment to LLM judges, which accept corrupted reasoning almost as often as clean reasoning [12]; and approach-level diversity has been measured by an LLM judge calibrated on 80 human-labelled pairs, with no human judging RLVR outputs [13]. To our knowledge, no study has asked people what RLVR actually loses: whether the solutions that disappear are distinct strategies, and whether the problems that leave the boundary were ever solved by valid reasoning.

A third gap concerns the fixes. *Intra-group* methods give rare correct modes of one prompt more credit than common ones [14]–[19]; *cross-prompt* methods reweight whole problems or tasks [3], [10], [20], [21]. They act on different axes but have never been combined and tested for complementarity (ReCo combines two corrections, both within one prompt [15]). We address all three gaps in one pre-registered study:

- *Is shrinkage real at SLM scale?* A multi-seed measurement on Qwen2.5-0.5B-Instruct [22] trained on MATH [23], with problem-level boundary entries and exits and the number of rollouts per training problem [10].

- *What is lost, according to people?* A blinded human audit of (H1) whether the solution modes our partitioner finds are distinct strategies and (H2) whether the base model’s successes on problems lost to RLVR were valid reasoning, against difficulty-matched retained problems.
- *Do the fixes compose?* A 2×3 factorial crossing a topic-level budget on positive advantage mass (CB, macro) with no micro term, the per-prompt SELF filter, or intra-group mode-rarity redistribution (MEG), with seed-paired primary contrasts and Holm correction.
- *Evidence that at 0.5B and about two rollouts per training problem RLVR expands the boundary slightly instead of shrinking it, that neither level of fix nor their composite moves the shrinkage slope by more than about one point per unit $\ln k$, and that correct-solution diversity does not track coverage across runs.*^[R2]

II. RELATED WORK

Is shrinkage real? Yue et al. [2] showed that RLVR-trained models are overtaken by their base models at large k . Nguyen et al. [3] attribute this to negative interference and winner-take-all reinforcement and propose SELF, which trains only on problems whose greedy answer is wrong; Wu and Xuan [5] argue that RLVR cannot leave the base model’s support, Zhou and Li [6] locate the contraction at the first decision of a solution, and Zhou [4] diagnoses the inversion. Against this, ProRL [7] reports expansion after prolonged, KL-controlled training, Wen et al. [8] report expansion under CoT-Pass@ k , and Yao et al. [9] reconcile the two with an exploitation stage that shrinks and an exploration stage that expands. Yuan et al. [10] relate the decline to few rollouts per problem and gate problems by success count (BBG), and Dragoi et al. [11] show that Pass@ k is biased toward lucky hits. Other responses change the optimizer or the search [24]–[26]. At SLM scale, GRPO dynamics [27], difficulty scaling [28] and reward granularity [29] have been studied, but not shrinkage.

Measuring what is lost. Lee et al. [13] show that embedding similarity tracks phrasing more than strategy, that word-bigram overlap tracks approach better, and that only 0.7% of GSM8K problems admit more than one approach against 33% on MATH; their RLVR results come from an LLM judge calibrated on 80 human-labelled pairs. Yang et al. [30] code the strategies of frontier models with AI coders and human adjudication, without RL. Taşaltı et al. [12] find that CoT-Pass@ k judges accept corrupted chains, and that Pass@ k exceeds CoT-Pass@ k by 19.7 points on Qwen2.5-generation solvers. Our audit puts people directly on RLVR outcomes.

Intra-group credit redistribution. Cue-GRPO [14] redistributes positive credit over a partition of the correct rollouts by rarity, leaving incorrect rollouts untouched and restoring the mean; we adopt its rule as our micro term. ReCo [15] reweights against distributional concentration at the response and token level; GCPO [16], DRA-GRPO [17], EDAS [18] and TreeAdv [19] reshape advantages using diversity or structure within the group.

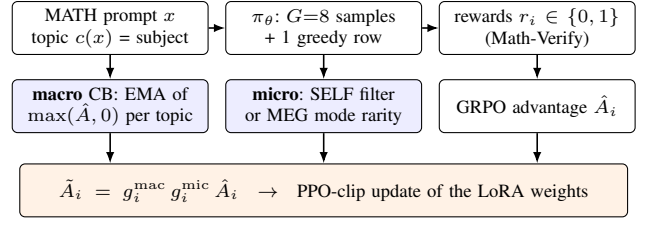


Fig. 1. Gated GRPO. The macro gate sees which topic a prompt belongs to and how much positive advantage that topic has absorbed in past steps; the micro gate sees the rollouts (and greedy answer) of the current group. Both scale the same advantage.

TABLE I
THE 2×3 FACTORIAL. COMPOSITES MULTIPLY THE MACRO AND MICRO GATES.

macro factor	micro factor		
	none	SELF filter	MEG redistribution
none	VANILLA	SELF	MEG
CB topic budget	CB	CB×SELF	CB×MEG (HEG-GRPO)

Cross-prompt reweighting. MT-GRPO [21] reweights explicit tasks for worst-task reliability; CurveRL [20] reweights prompts by their pass-rate distribution; BBG gates problems by success count [10]; SELF selects by greedy failure [3]. CB differs in granularity (latent topics rather than tasks or single prompts), in signal (each topic’s share of positive advantage mass), and in mechanism (a stateful multiplicative gate). Our question is not which gate is best but whether a topic-level and an intra-group intervention compose.

III. METHOD

A. GRPO with gated advantages

For a prompt x the policy π_θ samples a group of G completions $y_1, \dots, y_G$ with binary rewards $r_i \in \{0, 1\}$ (answer correctness). GRPO [1] uses the group-normalized advantage

$$\hat{A}_i = \frac{r_i - \text{mean}(r)}{\text{std}(r) + \epsilon}. \quad (1)$$

Every condition in this study multiplies it by a non-negative gate weight, $\tilde{A}_i = g_i \hat{A}_i$ with $g_i = g_i^{\text{mac}} g_i^{\text{mic}}$, and minimizes the PPO-clip loss [31], averaged over all T completion tokens in the batch [32]:

$$\mathcal{L}(\theta) = -\frac{1}{T} \sum_i \sum_{t=1}^{|y_i|} \min(\rho_{i,t} \tilde{A}_i, \text{clip}(\rho_{i,t}, 1 - \epsilon, 1 + \epsilon) \tilde{A}_i), \quad (2)$$

where $\rho_{i,t}$ is the token probability ratio. Updates are on-policy and use no KL penalty. Vanilla GRPO is $g_i \equiv 1$. Fig. 1 summarizes the pipeline.

B. Macro gate: topic budget (CB)

Topics are the $C = 7$ MATH subjects. After each step, CB updates an exponential moving average of each topic’s *spend*, the mean positive advantage mass of its rollouts in that step, $s_c \leftarrow (1 - \eta) s_c + \eta \text{mean}_{i:c(x_i)=c} \max(\tilde{A}_i, 0)$, with $\eta =$

0.05. Spend counts only positive mass, so corrective negative gradient never uses up a topic’s budget. With the overspend ratio $\rho_c = s_c / \frac{1}{C} \sum_{c'} s_{c'}$, the gate is

$$\tilde{g}_c = \delta^{\max(0, \rho_c - \vartheta)}, \quad \vartheta = 1.1, \delta = 0.18, \quad (3)$$

so a topic at $1.5\times$ the average spend gets half the credit of an in-budget one ($\delta^{0.4} = 0.5$). The gate acts only on rollouts with positive advantage. Their weights $\tilde{g}_{c(x_i)}$ are clipped to $[0.3, 3]$ and rescaled to mean 1 over the step, so CB moves positive credit from over-spending topics to the others rather than lowering the effective learning rate; negative advantages are left alone. The original specification ($\vartheta = 1.5$, $\delta = 0.98$, down-weighting only) never fired in the pilot (spend ratios stayed within $[0.38, 1.46]$), so we changed it before any main run; the pre-registration records this.

C. Micro gates: SELF filter and mode-rarity redistribution (MEG)

SELF. Following [3], a prompt whose greedy answer is already correct gets $g_x^{\text{self}} = \lambda$ ($\lambda = 0$, a hard filter), otherwise 1. The greedy answer is decoded as one extra row of the same generation call (its logits are pinned to the argmax), so it costs one sequence per prompt. We implement the selection rule only, not SELF’s forward-KL term.

MEG. Let $\mathcal{C}$ be the N correct rollouts of a group, partitioned into solution modes, and let $|M(i)|$ be the size of the mode containing rollout i . Following Cue-GRPO [14],

$$w_i = \frac{N |M(i)|^{-\alpha}}{\sum_{j \in \mathcal{C}} |M(j)|^{-\alpha}} \quad (i \in \mathcal{C}), \quad w_i = 1 \quad (i \notin \mathcal{C}), \quad (4)$$

with $\alpha = 0.8$; weights are clipped to $[0.3, 3]$ and rescaled to mean 1 over $\mathcal{C}$. Because the mean is preserved, MEG changes *which* correct solutions receive credit, not how much; an earlier down-weight-only version was confounded with a lower learning rate. We call this term the mode-equalizing gate.

Mode partitioner. Modes come from agglomerative clustering of the correct completions, either on MiniLM [33] sentence embeddings [34] (cosine threshold) or on word-bigram Jaccard distance. Because embeddings track phrasing more than strategy [13], the partitioner is not hand-tuned: a pre-registered rule, applied to base-model rollouts before training, picks the candidate whose median number of modes per group lies in $[2, 4]$ and is closest to 3 (ties go to bigram). It selected word-bigram Jaccard, threshold 0.4 (median 3.0 modes; 90% of groups with more than one mode; Table II). The rejected embedding candidates illustrate the risk: at the thresholds commonly used, *they place almost every group’s correct rollouts in a single mode*^[R15], which would turn MEG into a no-op.

D. Factorial design

Table I lists the six conditions. Composites multiply the two gates.

TABLE II
PARTITIONER CALIBRATION ON BASE-MODEL ROLLOUTS (GROUPS WITH ≥ 4 CORRECT). RULE: MEDIAN MODES IN $[2, 4]$, CLOSEST TO 3; $\star =$ SELECTED.

MiniLM cosine τ	0.80	0.85	0.90	0.95	
median modes	1.0	1.0	1.0	3.5	
multi-mode groups (%)	2	12	48	88	
Word-bigram threshold	0.30	0.40	0.50	0.60	0.70
median modes	5.0	3.0 $\star$	2.0	1.0	1.0
multi-mode groups (%)	100	90	60	35	5

TABLE III
TRAINING AND EVALUATION SETTINGS, IDENTICAL FOR ALL CONDITIONS.

Model	Qwen2.5-0.5B-Instruct, bf16 weights
Adapters	LoRA $r=16$, $\alpha=32$, dropout 0, all projections
Optimizer	AdamW, lr 1×10^{-5} , cosine, 10% warm-up, grad-norm clip 1.0
Batch	16 prompts $\times G=8$ rollouts = 128 rollouts per step
Steps	120; MATH levels 1–5
Sampling	$T=0.7$, top- p 0.95, no top- k , $\leq 1,024$ new tokens
Loss	PPO-clip $\varepsilon=0.2$, on-policy, token-level mean, no KL
CB / SELF / MEG	$\eta=0.05$, $\vartheta=1.1$, $\delta=0.18$ / $\lambda=0$ / $\alpha=0.8$, clip $[0.3, 3]$
Evaluation	$n=32$ samples, $T=0.6$, top- p 0.95; MATH-500, 1,024 tokens
Seeds	3 per condition
Hardware	NVIDIA A10G 24 GB (AWS g5.xlarge), one run per GPU, two GPUs in parallel

IV. EXPERIMENTAL SETUP

Training. We train Qwen2.5-0.5B-Instruct [22] with LoRA [35] adapters on the MATH training split [23] (7,496 problems with a boxed answer), prompted with the standard “reason step by step and put the final answer in `\boxed{\hspace{1cm}}`” instruction; Table III lists every setting. Rewards are binary correctness checked with Math-Verify [36] under a time limit, with a guarded symbolic fallback; a strict tag-format reward was dropped after it proved identically zero, hence inert under group normalization. The budget amounts to 0.26 epochs, or 2.0 rollouts per training problem. Seed s fixes the data order and LoRA initialization across conditions, so conditions are compared in seed-paired blocks.

Evaluation. Every model, including the base model, is scored on the same problems: all of MATH-500 [37] (1,024 tokens), with $n = 32$ samples per problem at temperature 0.6 and top- p 0.95. Pass@ k for $k \in \{1, 2, 4, 8, 16, 32\}$ uses the unbiased estimator [38] $1 - \binom{n-c}{k} / \binom{n}{k}$ for c correct samples. MATH-500 shares no problem statement with the training split; 7 of its problems (1.4%) are template siblings of training problems with a different question or numbers.

Outcome. The pre-registered primary outcome is the *shrinkage slope*: the least-squares slope of $\Delta(k) = \text{Pass}@k(\text{run}) - \text{Pass}@k(\text{base})$ against $\ln k$, reported in percentage points per unit $\ln k$. A negative slope means gains at small k turn into losses at large k ; higher means less shrinkage. We also count *boundary exits* (problems the base model solves at least once in 32 samples and the trained model never does) and *entries*

(the reverse).

Pre-registered analysis. All contrasts are linear combinations of condition slopes, paired by seed. The two primary contrasts are P1, $\text{CB} \times \text{MEG} - \text{CB} \times \text{SELF}$ (intra-group redistribution vs. per-prompt filtering, both under the budget), and P2, $\text{CB} \times \text{MEG} - \text{MEG}$ (does the macro level add to the micro level?). Secondary contrasts are the two macro $\times$ micro interactions (S1 for MEG, S2 for SELF), reported as 95% bounds rather than verdicts, and the macro main effect (S3); E1 is exploratory. Each contrast gets a t -interval over seeds and a hierarchical bootstrap that resamples seeds and problems jointly (2,000 resamples). An effect is claimed only if its Holm-adjusted [39] bootstrap $p < 0.05$ and its bootstrap interval excludes zero. Diagnostics, logged for every condition: normalized mode entropy of the correct rollouts in each group, the mean gate weight on positive-advantage rollouts, and the share of prompts SELF selects. The pre-registration also fixes how outcomes are worded: a significant P1 or P2 means the composition helps beyond its parts; if neither is significant and the intervals are narrow, the result is a bounded null; if vanilla GRPO shows no shrinkage, that is itself reported and the contrasts are still given.

Human audit (pre-registered, secondary). Two annotators [TODO: background], blind to condition, group and partitioner output, label two sheets built by our release. *H1, partition validity:* 60 pairs of correct base-model solutions to the same training problem, half placed in one mode by the partitioner and half in different modes, labelled SAME or DIFFERENT approach under the approach-level definition of [13]. We report Cohen’s κ and the partitioner’s balanced accuracy against the annotators’ consensus, and read the modes as approach-level only if it reaches 70%. *H2, what is lost:* MATH-500 problems the base model solves at least once in 32 samples but vanilla GRPO never solves in a majority of seeds (*exited*), each matched to a retained problem with the closest base solve count. For one random correct base-model solution per problem, annotators judge the reasoning VALID or FLAWED (right answer, unsupported reasoning). Rates are compared with a two-sided Fisher exact test; the pre-registered reading is “mostly lucky hits” if the exited rate is below 50% and significantly below the retained rate, “genuine loss” if it is at least 50% and not significantly lower, “mixed” otherwise.

Reproducibility. Code, logs, annotation sheets and the script behind every number are at github.com/2005legend/Reasoning-boundary-paradox-of-SLM. The pre-registration (written 2026-10-03, changed only by dated entries) was first committed after the runs (4cdc43a), so its dates are self-reported.

V. RESULTS

A. Shrinkage at 0.5B

Vanilla GRPO does not reproduce the inversion at this budget.^[R3] Pass@1 moves from 30.3% to 30.4% and Pass@32 rises from 73.0% to 75.3% (Table IV); the curve stays above zero up to $k = 32$ and its slope is positive (+0.66; Fig. 2, left). On average 36.3 problems enter the boundary and 25.0 leave it.

TABLE IV

MATH-500 PASS@ k (%), MEAN OVER SEEDS), SHRINKAGE SLOPE (PP PER UNIT $\ln k$; HIGHER IS BETTER) AND BOUNDARY ENTRIES/EXITS (OF 500 PROBLEMS).

Condition	P@1	P@4	P@16	P@32	Slope	In / Out
Base model	30.3	50.1	66.3	73.0	–	–
VANILLA	30.4	51.1	68.4	75.3	+0.66	36.3 / 25.0
SELF	30.8	51.1	68.0	75.2	+0.48	36.5 / 25.5
MEG	30.4	50.8	67.1	73.5	+0.13	30.7 / 28.3
CB	31.2	51.3	67.9	74.6	+0.21	34.0 / 26.0
CB $\times$ SELF	31.4	51.6	68.5	75.3	+0.37	32.3 / 20.7
CB $\times$ MEG	30.8	51.3	68.1	74.7	+0.35	33.3 / 25.0

TABLE V

CONTRASTS ON THE MATH-500 SHRINKAGE SLOPE (PP PER UNIT $\ln k$), PAIRED BY SEED. CI: HIERARCHICAL BOOTSTRAP OVER SEEDS AND PROBLEMS. p : HOLM-ADJUSTED FOR P1–P2, UNADJUSTED OTHERWISE.

Contrast	Est.	95% CI	p
P1 CB $\times$ MEG – CB $\times$ SELF	−0.02	[−0.79, +0.68]	1.000
P2 CB $\times$ MEG – MEG	+0.22	[−0.60, +1.05]	1.000
S1 CB $\times$ MEG interaction	+0.38	[−0.94, +1.69]	0.493
S2 CB $\times$ SELF interaction	+0.46	[−0.71, +1.68]	0.396
S3 CB main effect	−0.18	[−0.59, +0.25]	0.360
E1 CB $\times$ MEG – Vanilla	−0.31	[−0.91, +0.30]	0.291

Every other condition shows the same pattern, a positive slope and more entries than exits. Each training problem received 2.0 rollouts on average, and aggregate shrinkage has been tied to how often each problem is revisited [10]; our runs sit at the low-visit end that SLM budgets allow.

B. Pre-registered contrasts

All six slopes are positive, from +0.13 (MEG) to +0.66 (VANILLA), and no gate beats vanilla^[R4] (Table IV). Neither primary contrast clears the pre-registered bar (Table V): P1 = −0.02 (CI [−0.79, +0.68], Holm $p = 1.000$) and P2 = +0.22 (CI [−0.60, +1.05], Holm $p = 1.000$).^[R5] In units a reader can hold, multiplying by $\ln 32$, the two primary intervals correspond to [−2.7, +2.4] and [−2.1, +3.6] points of gain between $k = 1$ and $k = 32$: any difference between the micro terms is smaller than about three points. The interactions are wider, [−3.2, +5.9] for CB $\times$ MEG and [−2.5, +5.8] for CB $\times$ SELF, which admits effects larger than vanilla’s whole gain at $k = 32$, so with two seeds per cell the interaction question stays open. The macro main effect is −0.18 (CI [−0.59, +0.25]), and against VANILLA the composite changes the slope by −0.31 (CI [−0.91, +0.30]).^[R6] Following the pre-registration, we read the primary contrasts as a bounded null. The gates also cost reproducibility: across seeds the slope’s standard deviation is 0.04 for VANILLA and 0.40 for MEG (0.25 and 0.28 for the composites), a descriptive contrast at three seeds.

C. What each level does

Mode entropy of the correct rollouts starts at 0.63 and does not decay under vanilla GRPO^[R7] (Fig. 2, right). Its

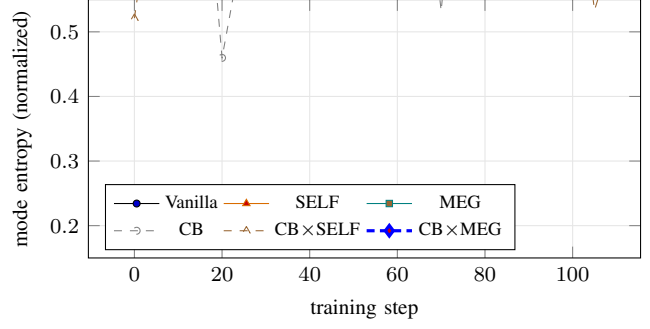
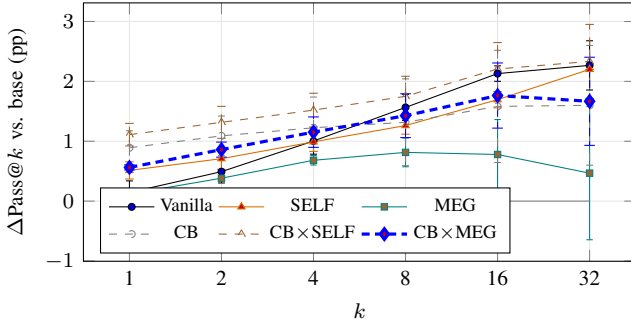


Fig. 2. Left: change in MATH-500 Pass@ k relative to the base model (mean $\pm$ s.d. over seeds); a curve that falls below zero at large k has shrunk the reasoning boundary. Right: normalized entropy of the solution-mode distribution among correct rollouts during training (mean over seeds).

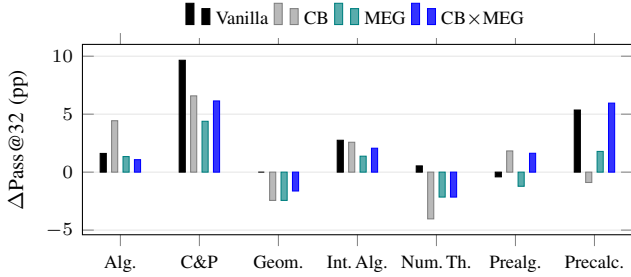


Fig. 3. Change in MATH-500 Pass@32 relative to the base model by subject, i.e. by the macro gate’s topics (mean over seeds).

training mean is 0.75 for VANILLA and 0.73 for CB×MEG. At evaluation, vanilla’s correct answers to a problem fall into 7.39 solution modes on average, against 6.73 for the base model, and its largest mode holds 39% of them against 43%. There is no collapse for MEG to undo. SELF skips the 59% of prompts it already solves, keeping a mean positive weight of 0.29. Changes are uneven across subjects (Fig. 3): under vanilla GRPO, Pass@32 rises by +9.6 pp in Counting & Probability but +0.0 pp in Geometry. The macro gate binds but changes nothing: in the CB runs a subject’s share of positive credit does not predict its Pass@32 change (Spearman $\rho = +0.01$, $n = 28$ subject-runs), so the premise of a topic budget finds no support here.^[R8]

Boundary exits do not separate the conditions much.^[R9] CB×SELF loses the fewest problems (20.7) and MEG the most (28.3), against 25.0 for VANILLA. Across all 16 runs, runs that gained more distinct correct modes lost slightly more problems (Spearman $\rho = +0.57$, $p = 0.02$), but over the 7 correlations we computed this does not survive correction (Benjamini–Hochberg $p = 0.16$), and none of the others comes close. Since Section V-D shows these modes are mostly wording, we treat the correlation as a hypothesis, not a finding.

D. What is lost: the human audit

H1, are the modes approaches? The annotators called 58 and 51 of the 60 pairs the same approach, although the partitioner had split half of them into different modes. They agreed on 51 pairs (85%): 50 SAME and 1 DIFFERENT. Cohen’s κ

is only 0.13 because almost all labels fall in one class, which depresses κ however well raters agree; the prevalence-adjusted PABAK is +0.70. Against the consensus the partitioner scores 55%, below the 98% of a labeller that always answers SAME: it splits pairs that people see as one approach. The pre-registered balanced accuracy, 77%, is formally above the 70% bar, but one of its classes holds a single pair, so we do not rely on it. We read the modes as surface clusters, and MEG as redistributing credit over them.^[R17]

H2, what is lost? Despite its net gains, vanilla GRPO lost 20 problems in a majority of seeds. The annotators did not agree on whether the base model’s successes on those problems were valid: A rated 11 of 40 solutions valid and B 28 ($\kappa = -0.06$, agreement on 15 items). On those items, 83% of exited solutions were valid (CI [44, 97], $n = 6$) against 22% of retained ones (CI [6, 55], $n = 9$; Fisher $p = 0.041$). The literal pre-registered rule reads this as a genuine loss, but a subset selected by agreement at chance-level reliability cannot carry that; each annotator alone shows the same direction without significance (A: $p = 0.480$; B: $p = 0.082$).^[R18] We report H2 as inconclusive.

VI. DISCUSSION AND LIMITATIONS

Training intensity, not size. With about two rollouts per training problem and a policy that moved little (final KL to the base model about an order of magnitude above its bf16 noise floor), RLVR on a 0.5B model expands the boundary slightly. This is what accounts that tie shrinkage to repeated reinforcement of saturated problems [10] or to a long exploitation phase [9] predict, and the inversions reported elsewhere [2], [3] come from far longer training. We therefore read our null as evidence about the training regime a single-GPU budget reaches, not about model size, and it suggests a direct test: raise rollouts per problem at 0.5B and shrinkage should appear. Fixes designed against shrinkage have little to act on in this regime.^[R13] **The partition is the weak link.** Embedding partitions put nearly every group into one mode (Table II); the bigram partition we selected finds about three modes that people see as one approach. Neither tracks what intra-group methods assume it tracks, which matters for any method that reweights over such partitions [14]–[18]. This is

compatible with Lee et al.’s finding that a third of MATH problems admit more than one approach [13]: the alternatives exist in the dataset but rarely in what a 0.5B model samples. We do not claim that shrinkage is absent at larger budgets; the bounds in Table V are what a longer study would need to beat.

Limitations. One model at one size; LoRA, one RL algorithm and short training (120 steps, cut from a planned 240), evaluated only on MATH-500. Two or three seeds per cell leave the interactions open. SELF omits its forward-KL term, and no random-partition control was run. The audit has two non-author annotators who differed in strictness on H2, and the pre-registered third annotator was not used. The correlations in Section V-C are descriptive and were added after the pre-registration.

VII. CONCLUSION

In a pre-registered study of Qwen2.5-0.5B-Instruct with a blinded human audit, **shrinkage does not appear at about two rollouts per training problem, and neither a topic-level budget, intra-group mode-rarity redistribution, nor their composite moves the shrinkage slope by more than about one point per unit $\ln k$ [R14].** People judged the partitioner’s modes to be wording variants of one approach. The protocol runs on a single 24 GB GPU in 86 GPU-hours and is released.

ACKNOWLEDGMENT

We thank the two annotators. **[TODO: AWS credits or other funding, if any.]** Compute: 86 A10G GPU-hours, about \$87. An AI assistant (Claude, Anthropic) helped write code and draft parts of the text; the authors verified the results and take responsibility for the content.

REFERENCES

- [1] Z. Shao *et al.*, “DeepSeekMath: Pushing the limits of mathematical reasoning in open language models,” *arXiv preprint arXiv:2402.03300*, 2024.
- [2] Y. Yue *et al.*, “Does reinforcement learning really incentivize reasoning capacity in LLMs beyond the base model?” *arXiv preprint arXiv:2504.13837*, 2025.
- [3] P. M. Nguyen, C. D. La, D. M. H. Nguyen, N. V. Chawla, B. T. Nguyen, and K. D. Doan, “The reasoning boundary paradox: How reinforcement learning constrains language models,” *arXiv preprint arXiv:2510.02230*, 2025.
- [4] T. Zhou, “When RLVR shrinks the reasoning boundary: Diagnosing Pass@k inversion,” *arXiv preprint arXiv:2607.20543*, 2026.
- [5] F. Wu *et al.*, “The invisible leash: Why RLVR may or may not escape its origin,” *arXiv preprint arXiv:2507.14843*, 2025.
- [6] Q. Zhou and R. Li, “Locked at the entrance, open inside: Where RLVR narrows the solution space,” *arXiv preprint arXiv:2608.29188*, 2026.
- [7] M. Liu *et al.*, “ProRL: Prolonged reinforcement learning expands reasoning boundaries in large language models,” *arXiv preprint arXiv:2505.24864*, 2025.
- [8] X. Wen *et al.*, “Reinforcement learning with verifiable rewards implicitly incentivizes correct reasoning in base LLMs,” *arXiv preprint arXiv:2506.14245*, 2025.
- [9] X. Yao *et al.*, “The debate on RLVR reasoning capability boundary: Shrinkage, expansion, or both? a two-stage dynamic view,” *arXiv preprint arXiv:2510.04028*, 2025.
- [10] S. Yuan *et al.*, “Understanding diversity collapse in RLVR via the lens of overtraining,” *arXiv preprint arXiv:2606.15455*, 2026.
- [11] M. Dragoi, I. Pintilie, F. Gogianu, and F. Brad, “Beyond Pass@k: Breadth-depth metrics for reasoning boundaries,” *arXiv preprint arXiv:2510.08325*, 2025.
- [12] T. T. Taşaltı, B. Hüdaverdi, and D. Semedo, “Does CoT-Pass@k really check the CoT? a multilingual mathematical audit,” *arXiv preprint arXiv:2609.32622*, 2026.
- [13] S. Lee, M. Kim, J. Kim, D. Kim, S. Rhee, and K. Jung, “Are we measuring strategy or phrasing? The gap between surface- and approach-level diversity in LLM math reasoning,” *arXiv preprint arXiv:2606.29985*, 2026.
- [14] Z. Cao, M. Wen, and F. Chen, “When correct solutions repeat: Rarity-aware credit redistribution for GRPO,” *arXiv preprint arXiv:2608.03467*, 2026.
- [15] J. Park, J. Hwang, W. Cho, and T. Kim, “ReCo: Reweighting GRPO against distributional concentration,” *arXiv preprint arXiv:2607.26862*, 2026.
- [16] H. Chen, T. Liang, W.-S. Zheng, and J.-F. Hu, “Breaking winner-takes-all: Cooperative policy optimization improves diverse LLM reasoning,” *arXiv preprint arXiv:2605.11461*, 2026.
- [17] X. Chen *et al.*, “DRA-GRPO: Your GRPO needs to know diverse reasoning paths for mathematical reasoning,” *arXiv preprint arXiv:2505.09655*, 2025.
- [18] W. Liu *et al.*, “Leveraging error diversity in group rollouts for reinforcement learning,” *arXiv preprint arXiv:2605.17333*, 2026.
- [19] L. Cao *et al.*, “TreeAdv: Tree-structured advantage redistribution for group-based RL,” *arXiv preprint arXiv:2601.03703*, 2026.
- [20] K. Sun, Y. Zhao, J. Xin, Q. Long, and W. Su, “CurveRL: Principled distribution-aware context reweighting for LLM reasoning,” *arXiv preprint arXiv:2605.24331*, 2026.
- [21] S. S. Ramesh *et al.*, “Multi-task GRPO: Reliable LLM reasoning across tasks,” *arXiv preprint arXiv:2602.05547*, 2026.
- [22] Qwen Team, “Qwen2.5 technical report,” *arXiv preprint arXiv:2412.15115*, 2024.
- [23] D. Hendrycks *et al.*, “Measuring mathematical problem solving with the MATH dataset,” *arXiv preprint arXiv:2103.03874*, 2021.
- [24] Y. Ba *et al.*, “Understanding evolution strategies for LLM reasoning: Broader reasoning coverage than GRPO,” *arXiv preprint arXiv:2608.27351*, 2026.
- [25] Y. Yu, S. Jang, and H. Yu, “Difficulty-adaptive tree-structured policy optimization for expanding reasoning coverage in RLVR,” *arXiv preprint arXiv:2609.08650*, 2026.
- [26] J. Decugis, S. O’Brien, F. Bach, G. Synnaeve, and T. Cohen, “Don’t let gains FADE: Breaking down policy gradient weights in RL,” *arXiv preprint arXiv:2607.01490*, 2026.
- [27] R. Ghosh, V. Bhargava, H. Wong, A. Singhal, and D. Dutta, “GRPO training dynamics for small language models,” *arXiv preprint arXiv:2609.39321*, 2026.
- [28] S. Yadav, S. Yadav, and P. Goyal, “Limits of difficulty scaling: Hard samples yield diminishing returns in GRPO-tuned SLMs,” *arXiv preprint arXiv:2604.06298*, 2026.
- [29] A. R. Landy, R. Glick, and O. Kaul, “Reward granularity in RLVR: Comparing process and outcome reward structures for mathematical reasoning in small language models,” *arXiv preprint arXiv:2607.02869*, 2026.
- [30] X. Yang, X. Zhang, H. Hu, and F. Ji, “Beyond accuracy: Evaluating strategy diversity in LLM mathematical reasoning,” *arXiv preprint arXiv:2605.09292*, 2026.
- [31] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal policy optimization algorithms,” *arXiv preprint arXiv:1707.06347*, 2017.
- [32] Q. Yu *et al.*, “DAPO: An open-source LLM reinforcement learning system at scale,” *arXiv preprint arXiv:2503.14476*, 2025.
- [33] W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, and M. Zhou, “MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers,” *arXiv preprint arXiv:2002.10957*, 2020.
- [34] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence embeddings using Siamese BERT-networks,” *arXiv preprint arXiv:1908.10084*, 2019.
- [35] E. J. Hu *et al.*, “LoRA: Low-rank adaptation of large language models,” *arXiv preprint arXiv:2106.09685*, 2021.
- [36] Hugging Face, “Math-Verify: Math verification library,” <https://github.com/huggingface/Math-Verify>, 2025, version 0.9.0.
- [37] H. Lightman *et al.*, “Let’s verify step by step,” *arXiv preprint arXiv:2305.20050*, 2023.
- [38] M. Chen *et al.*, “Evaluating large language models trained on code,” *arXiv preprint arXiv:2107.03374*, 2021.
- [39] S. Holm, “A simple sequentially rejective multiple test procedure,” *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.